Reduced Optimal Control with PDE Constraints

Model, Discretization, and Optimization Methods

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Fachseminar

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Outline

- Problem formulation
- Variational formulation and discretization
- Reduced problem: gradients and optimality
- Optimization methods (GD, BB, Nesterov, CG)
- Numerical results and mesh-independence

Problem formulation

Goal: Find a control u such that the state y matches a desired profile y_d.

PDE constraint:

$$-\Delta y = \chi_\omega\, u \quad \text{in } \Omega, \qquad y = 0 \quad \text{on } \partial\Omega.$$

Cost:

$$J(y,u) = \tfrac{1}{2}\|y - y_d\|^2_{L^2(\Omega)} + \tfrac{\beta}{2}\|u\|^2_{L^2(\Omega)}, \qquad \beta > 0.$$

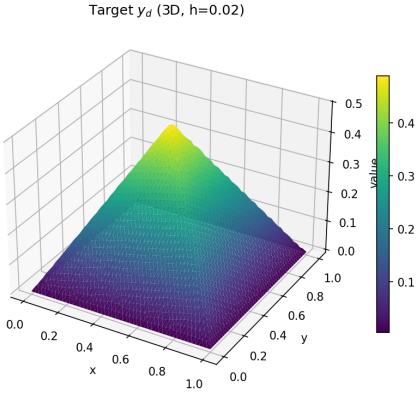
- $\Omega = [0,1]^2$, control acts only on a subdomain $\omega \subset \Omega$.
- Spaces: $V = H_0^1(\Omega)$, $U = L^2(\Omega)$.

In our computations, ω is the disk

$$\omega := \{x \in \Omega : (x_1 - 0.5)^2 + (x_2 - 0.5)^2 \le 0.1^2\}.$$

Target potential y_d

$$y_d(x) = 0.5 - \max\left(|x_1 - 0.5|,\ |x_2 - 0.5|\right).$$



Variational formulation

Define

$$a(y, v) = \int_\Omega \nabla y \cdot \nabla v \, dx \quad \text{for } y, v \in V,$$
$$b(u, v) = \int_\omega u \, v \, dx \quad \text{for } u \in U, \ v \in V.$$

State: given $u \in U$, find $y \in V$ such that

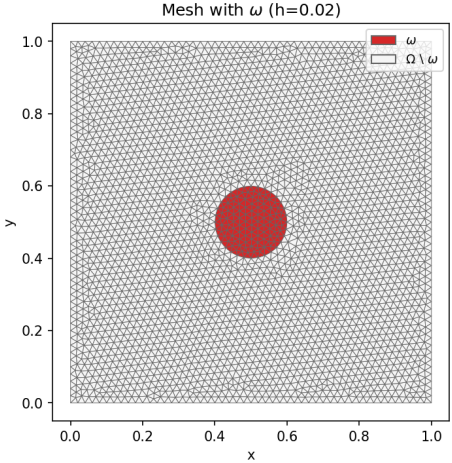
$$a(y, v) = b(u, v) \quad \forall v \in V.$$

Objective:

$$J(y, u) = \frac{1}{2} \int_\Omega (y - y_d)^2 \, dx + \frac{\beta}{2} \int_\Omega u^2 \, dx.$$

Discretization and discrete objective

Let $V_h \subset H_0^1(\Omega)$ be the conforming P1 space with basis $(\varphi_i)_{i=1}^n$.



Triangulation with subdomain ω.

Discretization and discrete objective

Assemble

$$A_{ij} = \int_\Omega \nabla\varphi_j \cdot \nabla\varphi_i \, dx, \qquad M_{ij} = \int_\Omega \varphi_j \varphi_i \, dx,$$

$$B_{ij} = \int_\omega \varphi_j \varphi_i \, dx, \qquad (M_U)_{ij} = \int_\Omega \varphi_j \varphi_i \, dx.$$

Discrete state:

$$A\boldsymbol{y} = B\boldsymbol{u}.$$

Discrete objective:

$$J(\boldsymbol{y}, \boldsymbol{u}) = \tfrac{1}{2}(\boldsymbol{y} - \boldsymbol{y}_d)^\top M(\boldsymbol{y} - \boldsymbol{y}_d) + \tfrac{\beta}{2}\boldsymbol{u}^\top M_U \boldsymbol{u}.$$

Reduced functional $F(\boldsymbol{u})$

Eliminate the state via the discrete PDE constraint

$$A\boldsymbol{y} = B\boldsymbol{u}, \qquad \boldsymbol{y}(\boldsymbol{u}) = A^{-1}B\boldsymbol{u}.$$

Insert $\boldsymbol{y}(\boldsymbol{u})$ into J to obtain the reduced objective

$$F(\boldsymbol{u}) = \tfrac{1}{2}\big(\boldsymbol{y}(\boldsymbol{u}) - \boldsymbol{y}_d\big)^\top M\big(\boldsymbol{y}(\boldsymbol{u}) - \boldsymbol{y}_d\big) + \tfrac{\beta}{2}\,\boldsymbol{u}^\top M_U \boldsymbol{u}.$$

U-gradient and Riesz map

We equip $U_h \cong \mathbb{R}^n$ with the mass-matrix inner product

$$\langle \boldsymbol{a}, \boldsymbol{b} \rangle_U := \boldsymbol{a}^\top M_U \boldsymbol{b}, \qquad \|\boldsymbol{a}\|_U := \sqrt{\boldsymbol{a}^\top M_U \boldsymbol{a}}.$$

The U-gradient $\nabla_U F(\boldsymbol{u})$ is defined by

$$\langle \nabla_U F(\boldsymbol{u}), \boldsymbol{v} \rangle_U = F'(\boldsymbol{u})[\boldsymbol{v}] \quad \forall \boldsymbol{v}.$$

If $\nabla F(\boldsymbol{u})$ denotes the coordinate gradient (Euclidean inner product), then

$$\nabla F(\boldsymbol{u}) = M_U \, \nabla_U F(\boldsymbol{u}).$$

Adjoint formula

Eliminate state: $\boldsymbol{y}(\boldsymbol{u}) = A^{-1}B\boldsymbol{u}$. Solve

$$A\boldsymbol{y} = B\boldsymbol{u}, \qquad A\boldsymbol{p} = M(\boldsymbol{y} - \boldsymbol{y}_d).$$

Then

$$\nabla F(\boldsymbol{u}) = B^\top \boldsymbol{p} + \beta M_U \boldsymbol{u}, \qquad \nabla_U F(\boldsymbol{u}) = M_U^{-1} B^\top \boldsymbol{p} + \beta \boldsymbol{u}.$$

Optimality condition

The reduced functional $F(\boldsymbol{u})$ is strictly convex, hence has a unique minimizer $\boldsymbol{u}_*$. The first-order condition is

$$\nabla_U F(\boldsymbol{u}_*) = \mathbf{0}.$$

Equivalently (multiply by M_U),

$$\nabla F(\boldsymbol{u}_*) = \mathbf{0}.$$

Convex quadratic problem formulation

Expand F as an unconstrained quadratic problem

$$F(\boldsymbol{u}) = \tfrac{1}{2}\boldsymbol{u}^\top Q\boldsymbol{u} - \boldsymbol{b}^\top \boldsymbol{u} + \text{const},$$

with

$$Q = B^\top A^{-1} M A^{-1} B + \beta M_U \quad \text{(SPD)}, \qquad \boldsymbol{b} = B^\top A^{-1} M \boldsymbol{y}_d.$$

Stationarity system:

$$Q\boldsymbol{u}_* = \boldsymbol{b}.$$

Lipschitz constant L and strong convexity m

In the U-inner product, the reduced Hessian is

$$H = M_U^{-1} Q \quad \text{(self-adjoint and SPD in } \langle \cdot, \cdot \rangle_U\text{)}.$$

Its extremal eigenvalues satisfy the generalized eigenproblem

$$Q\boldsymbol{v} = \lambda\, M_U \boldsymbol{v}.$$

Hence

$$L = \lambda_{\max}, \qquad m = \lambda_{\min}, \qquad \kappa = \frac{L}{m}.$$

These constants define the fixed GD step $\alpha = 1/L$ and Nesterov parameters.

Gradient descent (GD)

Iteration (fixed step):

$$\boldsymbol{u}_{k+1} = \boldsymbol{u}_k - \frac{1}{L}\,\nabla_U F(\boldsymbol{u}_k).$$

Stopping criterion:

$$\|\nabla_U F(\boldsymbol{u}_k)\|_U \leq \texttt{tol}_{abs} + \texttt{tol}_{rel}\,\|\nabla_U F(\mathbf{0})\|_U,$$

with $\texttt{tol}_{abs} = 10^{-5}$ and $\texttt{tol}_{rel} = 10^{-3}$.

Barzilai–Borwein (BB)

Idea: Use a secant-based scalar step size in the U-metric.

With $\boldsymbol{s}_k = \boldsymbol{u}_k - \boldsymbol{u}_{k-1}$ and $\boldsymbol{d}_k = \nabla_U F(\boldsymbol{u}_k) - \nabla_U F(\boldsymbol{u}_{k-1})$,

$$\alpha_k^{\mathrm{BB1}} = \frac{\langle \boldsymbol{s}_k, \boldsymbol{s}_k \rangle_U}{\langle \boldsymbol{s}_k, \boldsymbol{d}_k \rangle_U}, \qquad \alpha_k^{\mathrm{BB2}} = \frac{\langle \boldsymbol{s}_k, \boldsymbol{d}_k \rangle_U}{\langle \boldsymbol{d}_k, \boldsymbol{d}_k \rangle_U}.$$

Update: $\boldsymbol{u}_{k+1} = \boldsymbol{u}_k - \alpha_k \nabla_U F(\boldsymbol{u}_k)$ (alternating BB1/BB2).

Stopping criterion: same $\|\nabla_U F\|_U$ rule as GD.

Nesterov acceleration

With $\kappa = L/m$, define

$$\beta_N = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}, \qquad \boldsymbol{y}_k = \boldsymbol{u}_k + \beta_N(\boldsymbol{u}_k - \boldsymbol{u}_{k-1}).$$

Update (fixed step):

$$\boldsymbol{u}_{k+1} = \boldsymbol{y}_k - \tfrac{1}{L}\,\nabla_U F(\boldsymbol{y}_k).$$

Stopping criterion: same $\|\nabla_U F\|_U$ rule as GD.

Conjugate gradients (CG) for the stationarity system

Solve the SPD linear system

$$Q\boldsymbol{u} = \boldsymbol{b}, \qquad \boldsymbol{b} = B^\top A^{-1} M \boldsymbol{y}_d.$$

Stopping criterion:

$$\|\boldsymbol{r}_k\|_2 \le \texttt{rtol}\, \|\boldsymbol{b}\|_2, \qquad \texttt{rtol} = 10^{-4}, \qquad \boldsymbol{r}_k = \boldsymbol{b} - Q\boldsymbol{u}_k.$$

Link to the U-gradient: for the quadratic problem,

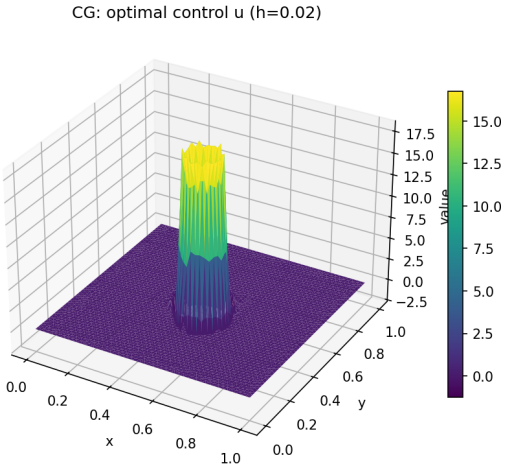
$$\nabla F(\boldsymbol{u}) = Q\boldsymbol{u} - \boldsymbol{b} = -\boldsymbol{r}, \qquad \nabla_U F(\boldsymbol{u}) = M_U^{-1}(Q\boldsymbol{u} - \boldsymbol{b}) = -M_U^{-1}\boldsymbol{r}.$$

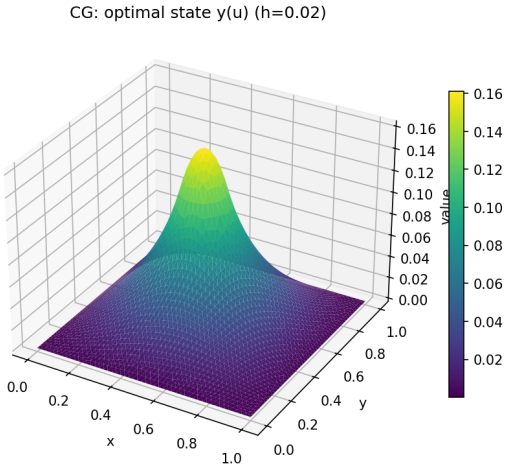
Thus

$$\|\nabla_U F(\boldsymbol{u}_k)\|_U = \sqrt{\boldsymbol{r}_k^\top M_U^{-1} \boldsymbol{r}_k}.$$

Setup and representative solution ($h = 0.02$)

- Parameters: $\beta = 10^{-3}$
- Mesh: representative run shown for $h = 0.02$.

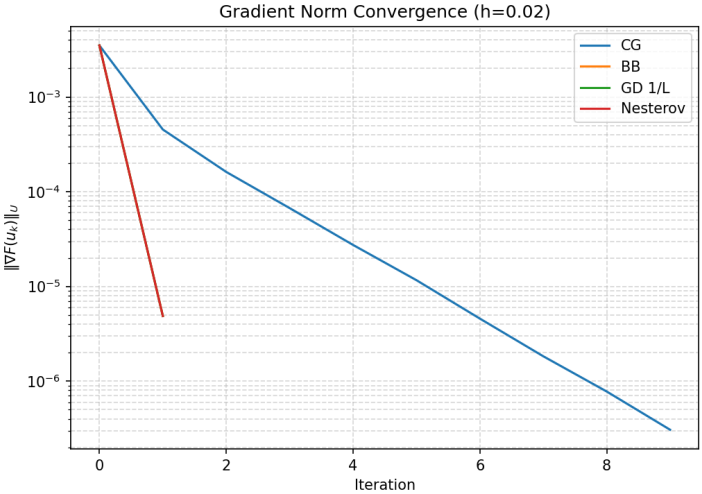




Numerical estimates for L and m

h	L_{est}	m_{est}	$\kappa = L/m$
0.06	1.2761×10^{-3}	1.0000×10^{-3}	1.276
0.04	1.2973×10^{-3}	1.0000×10^{-3}	1.297
0.03	1.3072×10^{-3}	1.0000×10^{-3}	1.307
0.02	1.3164×10^{-3}	1.0000×10^{-3}	1.316
0.015	1.3209×10^{-3}	1.0000×10^{-3}	1.321
0.012	1.3233×10^{-3}	1.0000×10^{-3}	1.323
0.01	1.3252×10^{-3}	1.0000×10^{-3}	1.325
0.008	1.3267×10^{-3}	1.0000×10^{-3}	1.327

Final cost, iterations, and gradient norm convergence ($h = 0.02$)



method	iters	$F(\boldsymbol{u}_{\text{final}})$
CG	9	$1.617\,08 \times 10^{-2}$
BB	1	$1.617\,09 \times 10^{-2}$
GD	1	$1.617\,09 \times 10^{-2}$
Nesterov	1	$1.617\,09 \times 10^{-2}$

Time measurements ($h = 0.02$)

Measured timings (seconds) for $h = 0.02$.

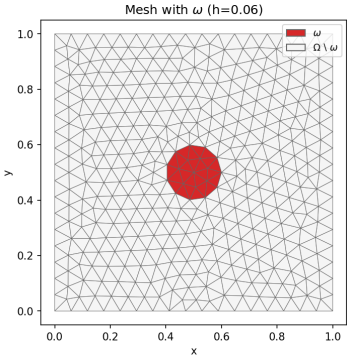
Method	t_L [s]	t_m [s]	t_{iter} [s]	t_{total} [s]
CG	0.0000	0.0000	0.0142	0.0142
BB	0.0000	0.0000	0.0068	0.0068
GD	0.0660	0.0000	0.0051	0.0710
Nesterov	0.0660	0.0822	0.0050	0.1532

Here $t_{\text{total}} = t_L + t_m + t_{\text{iter}}$.

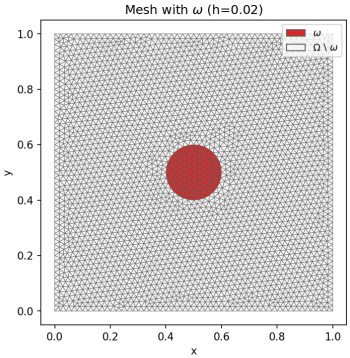
Interpretation

- The reduced problem is a strictly convex quadratic program, so all methods converge to the same minimizer.
- The gradient-norm histories converge in one iteration for BB, GD ($1/L$), and Nesterov.
- BB and CG avoid spectral estimates; on the representative mesh, BB achieves the smallest wall-clock time.

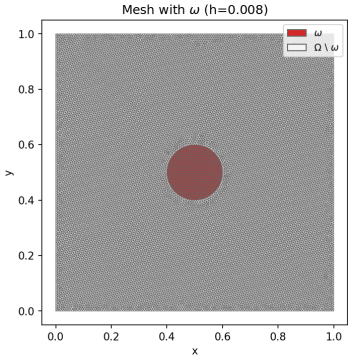
Mesh family (examples)



$h = 0.06$

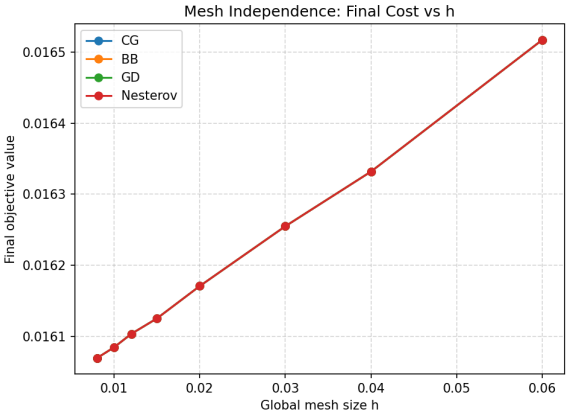


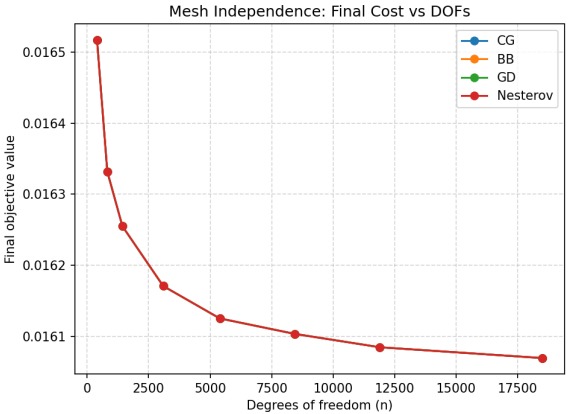
$h = 0.02$



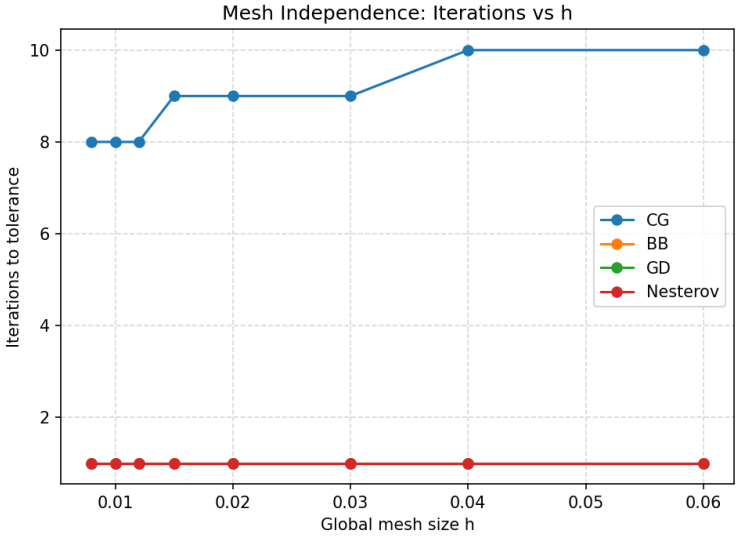
$h = 0.008$

Final cost under refinement

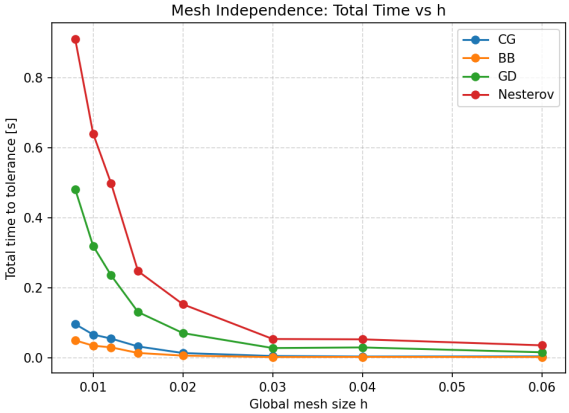


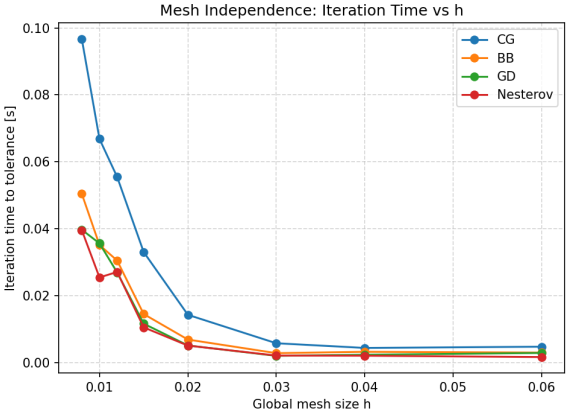


Iterations vs. h



Timings vs. h


Mesh Independence: Total Time vs h
Total time to tolerance [s]
Global mesh size h
CG
BB
GD
Nesterov
0.0
0.2
0.4
0.6
0.8
0.01
0.02
0.03
0.04
0.05
0.06


Mesh Independence: Iteration Time vs h
Iteration time to tolerance [s]
Global mesh size h
CG
BB
GD
Nesterov
0.00
0.02
0.04
0.06
0.08
0.10
0.01
0.02
0.03
0.04
0.05
0.06

Main findings

- The reduced problem is strictly convex quadratic, and all methods reach essentially the same final objective.
- Across the mesh family, objective values and norms are stable (mesh-independence trend).
- BB and CG do not require spectral estimates; GD/Nesterov pay one-time costs for L or (m, L).
- In these experiments, first-order methods converge in very few updates from $\boldsymbol{u}_0 = \nabla_U F(\mathbf{0})$.

Limitations and future work

- Unconstrained control (no box constraints such as $u_a \leq u \leq u_b$).
- Linear Poisson PDE only; no nonlinear state equations.
- No dedicated preconditioner study for CG / Krylov methods.

Possible next steps:

- Add control constraints and projection/semi-smooth Newton methods.
- Extend to nonlinear PDE constraints.
- Compare preconditioned Krylov solvers and matrix-free variants.

Acknowledgements / Tools

- Implementation and experimentation were carried out in Visual Studio Code.
- GitHub Copilot (free for students) was used as an AI assistant during development, in particular to speed up coding and help understand FEniCS-related implementation details.
- GitHub: github.com/MaxHartenberger

Backup: residual and U-gradient relation

For the quadratic reduced problem,

$$\nabla F(\boldsymbol{u}) = Q\boldsymbol{u} - \boldsymbol{b}, \qquad \boldsymbol{r} = \boldsymbol{b} - Q\boldsymbol{u} = -\nabla F(\boldsymbol{u}).$$

Hence

$$\nabla_U F(\boldsymbol{u}) = M_U^{-1}(Q\boldsymbol{u} - \boldsymbol{b}) = -M_U^{-1}\boldsymbol{r},$$

and therefore

$$\|\nabla_U F(\boldsymbol{u})\|_U = \sqrt{\boldsymbol{r}^\top M_U^{-1}\boldsymbol{r}}.$$

Iteration counting convention in plots/tables:

- BB/GD/Nesterov: number of performed updates
- CG: number of Krylov updates

Key settings used in all experiments

- Tikhonov parameter: $\beta = 10^{-3}$
- Control region: disk centered at $(0.5, 0.5)$ with radius 0.1
- Mesh family: $h \in \{0.06, 0.04, 0.03, 0.02, 0.015, 0.012, 0.01, 0.008\}$
- Initial iterate (all methods, including CG): $\boldsymbol{u}_0 = \nabla_U F(\mathbf{0})$
- Stopping rules:
  - BB/GD/Nesterov: $\|\nabla_U F(\boldsymbol{u}_k)\|_U \leq \texttt{tol}_{abs} + \texttt{tol}_{rel} \|\nabla_U F(\mathbf{0})\|_U$
  - CG: $\|\boldsymbol{r}_k\|_2 \leq \texttt{rtol}\, \|\boldsymbol{b}\|_2$, with $\boldsymbol{r}_k = \boldsymbol{b} - Q\boldsymbol{u}_k$